

Radar-Based Gait Analysis for Early Detection of Mobility Impairments

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Abstract—In the context of ageing societies and the increasing prevalence of neurological and musculoskeletal disorders, there is a growing need for unobtrusive methods to assess mobility impairments and fall risk outside of clinical environments. Conventional gait assessment techniques, such as the Timed Up and Go (TUG) test, are typically performed under clinical supervision or rely on wearable sensors, which can limit their practicality for continuous and long-term in-home monitoring. This paper presents a radar-based framework for contactless gait analysis aimed at assessing mobility impairments and fall risk within domestic environments. Human motion is recorded using a short-range radar sensor during a TUG-inspired protocol, and micro-Doppler signatures are analysed to extract spatiotemporal gait features. From these signatures, clinically relevant spatiotemporal gait parameters can be estimated with a high degree of accuracy using the proposed approach.

Index Terms—radar, gait analysis, micro-Doppler, spatiotemporal gait features, in-home gait analysis, assisted living

I. INTRODUCTION

The ageing process is associated with an increased likelihood of developing mobility limitations, which may arise from both the physiological decline of body systems and the presence of chronic neurological and musculoskeletal conditions such as Parkinson’s disease and arthritis. These limitations are further influenced by lifestyle and environmental factors, and can significantly affect an individual’s ability to live independently [1]. Functional mobility encompasses a range of everyday actions, including changing body posture, walking, turning, rising from a seated position, and navigating indoor environments, all of which are commonly assessed in clinical practice to evaluate overall mobility and fall risk.

Gait and mobility impairments are commonly evaluated using a range of established assessment techniques, including standardised clinical tests, laboratory-based motion capture systems, wearable sensing technologies, and vision-based approaches. Among clinical assessments, the Timed Up and Go (TUG) test [2] is widely adopted due to its simplicity and effectiveness in screening functional mobility and fall risk; however, it is typically conducted under clinical supervision and relies primarily on coarse timing-based metrics, providing only a snapshot of an individual’s mobility and remaining susceptible to inter-rater variability. Laboratory-based motion capture (MoCap) systems are often regarded as the gold

standard for gait analysis due to their ability to provide highly accurate kinematic measurements, yet their reliance on specialised infrastructure, reflective markers, controlled environments, and expert operation limits their accessibility and practicality for routine assessment or in-home use [3]. Wearable sensor-based approaches enable continuous monitoring but depend on correct sensor placement, regular maintenance, and sustained user compliance over long periods [4]. Vision-based systems can capture detailed motion information but raise privacy concerns and are sensitive to environmental factors such as lighting conditions, occlusions, and restricted fields of view [5]. Consequently, existing gait assessment technologies are not well suited for unobtrusive, long-term monitoring in domestic environments, underscoring the need for alternative sensing approaches that enable objective, privacy-preserving, and scalable in-home gait assessment.

Radar sensing has emerged as a promising modality for gait analysis due to its ability to operate in a contactless and privacy-preserving manner while maintaining robustness to environmental variations [6]. In contrast to vision-based systems, radar performance is largely independent of ambient lighting conditions and is less affected by moderate occlusions, which makes it well suited for deployment in domestic environments. Furthermore, radar sensors capture micro-Doppler signatures generated by the motion of different body segments, providing rich information on gait dynamics and enabling the extraction of spatio-temporal gait characteristics without the need for body-worn devices [7]. Recent advances in short-range radar technology have facilitated the development of compact, low-power, and cost-effective sensing platforms that can be readily integrated into home environments. Collectively, these properties position radar as a practical sensing approach for unobtrusive and long-term in-home gait monitoring, addressing key limitations associated with clinical, wearable, and vision-based assessment methods [8].

This paper proposes a radar-based gait analysis framework for the objective assessment of mobility impairments and fall-risk indicators in domestic environments. The framework extracts spatiotemporal gait features from radar micro-Doppler signatures acquired during a TUG-inspired protocol, enabling contactless and repeatable gait assessment. Motivated by the need for objective and scalable tools to support clinical

decision making in resource-constrained healthcare systems, such as the National Health Service (NHS) [9], the proposed approach aims to provide longitudinal in-home gait metrics. These metrics have the potential to support evidence-based prioritisation of clinical interventions, including total knee replacement surgery, based on functional mobility trends rather than isolated clinical assessments. The main contributions of this work include the development of the radar-based framework, the extraction of clinically relevant spatio-temporal gait features, and the demonstration of their sensitivity to mobility degradation and fall-risk indicators.

The remainder of this paper is organised as follows. Section II describes the proposed methodology, including the radar measurement setup, the TUG-inspired protocol, and the extraction of spatiotemporal gait features from radar micro-Doppler signatures. Section III presents the experimental results and discussion, examining the extracted gait features and their sensitivity to mobility impairments and fall-risk indicators. Finally, Section IV concludes the paper and outlines directions for future work, including extended in-home monitoring and potential clinical applications.

II. METHODOLOGY

A. TUG-Protocol

The TUG protocol is a standard clinical assessment used to evaluate functional mobility and fall risk. During the test, a subject rises from a seated position, walks a short distance (typically 3-6 m), turns, returns to the chair, and sits down again, with both completion time and movement quality providing indicators of mobility impairment and fall susceptibility [2]. In this work, a TUG-inspired protocol is employed to capture radar-based gait signatures associated with the walking and turning phases of the task within an in-home environment.

B. System Description and Dataset

The experimental data used in this study were obtained from the authors of prior work [10] and were collected from 50 participants using a frequency-modulated continuous-wave (FMCW) radar system (SDR-980AD2, Ancortek) operating in the X-band at 9.8 GHz with a bandwidth of 400 MHz. The radar transmits 1 ms chirps, resulting in a Doppler frequency range of approximately ± 500 Hz, which is suitable for indoor gait analysis. The system comprises one transmit and two receive antennas with a transmit power of +19 dBm. Signal downconversion is performed through hardware dechirping, and the resulting complex in-phase and quadrature (I/Q) signals are sampled at 128 kHz, yielding 128 samples per chirp. Each recording has a fixed duration of 10 s, which is sufficient for capturing the gait activities analysed in this work.

C. Radar Signal Processing

During each observation interval, the radar records sequences of FMCW chirps containing complex in-phase and quadrature (I/Q) samples. *Range processing* is first performed by applying a one-dimensional fast Fourier transform (FFT) along the fast-time dimension of each chirp to obtain range

profiles. For the n -th chirp, the resulting range spectrum is given by

$$X(r, n) = \sum_{m=0}^{M-1} x(m, n) e^{-j2\pi \frac{rm}{M}}, \quad (1)$$

where $x(m, n)$ denotes the complex baseband signal, m is the fast-time sample index, M is the number of samples per chirp, and r represents the range bin index. By stacking successive range profiles over slow time, *range-time maps* are formed, which are subsequently used for activity segmentation, including the identification of walking and turning phases within the TUG-inspired protocol.

Following segmentation, signal components associated with walking are selected for *micro-Doppler analysis*. A short-time Fourier transform (STFT) is applied to the range-time data matrix at the selected range bins using a Hamming window of 0.2 s duration with a 95% overlap, in order to obtain time-frequency representations of motion-induced Doppler variations. The STFT is expressed as

$$S(r, k, t) = \sum_{\tau} X(r, \tau) w(t - \tau) e^{-j2\pi k\tau}, \quad (2)$$

where $w(\cdot)$ denotes the window function, and k and t correspond to the Doppler frequency and slow-time indices, respectively. This processing yields micro-Doppler signatures with a range resolution of 37.5 cm, a maximum measurable velocity of 7.65 m/s, and a Doppler resolution of 1.25 Hz (corresponding to approximately 0.03 m/s). The resulting micro-Doppler signatures capture periodic limb motion and are subsequently used to extract spatio-temporal gait parameters for mobility and fall-risk assessment.

D. Region of Interest Extraction

Activity segmentation is a necessary step for extracting spatio-temporal gait parameters within the TUG protocol. Several approaches can be employed for this purpose [13], [14]; in this study, segmentation is performed using range-time maps. Specifically, the region of interest (ROI) corresponding to the subject's motion is identified on the range-time representations to isolate the relevant activity segments. The same ROI extraction procedure is subsequently applied to the micro-Doppler signatures in order to extract the envelope of the signatures, which forms the basis for subsequent spatio-temporal gait parameter estimation.

In this study, the adaptive thresholding algorithm proposed in [15], [16] is employed for region of interest extraction. This algorithm applies an adaptive threshold T to binarise the grayscale images and extract the ROI. The image is first converted into a grayscale image S with pixel intensities denoted by $I(x, y)$. Based on an initial threshold μ , the image is partitioned into two classes: pixels with intensities greater than μ and pixels with intensities less than or equal to μ . The threshold T is then iteratively updated using the statistical properties of the current window until convergence is achieved. Convergence is defined as the condition where the difference between μ and T falls below a predefined

parameter V , which is empirically set to 0.1 in this study to enable effective masking of radar images [16]. The adaptive thresholding algorithm used for region of interest extraction is summarised in Algorithm 1.

Algorithm 1 Adaptive Thresholding Algorithm

Initialization: $I(x, y)$ - input grayscale image S with N pixels. p_1, p_2 - subregions within image S , where p_1 and p_2 have N_1 and N_2 pixels, respectively. $b(x, y)$ - binary output for pixel (x, y) .

Set T - adaptive threshold value. V - specific parameter determining the threshold.

$$\mu \leftarrow \frac{1}{N} \sum_{I(x,y) \in S} I(x, y)$$

$$p_1(x, y) \leftarrow I(x, y) > \mu, p_2(x, y) \leftarrow I(x, y) \leq \mu$$

$$T \leftarrow \frac{1}{2} \left(\frac{1}{N_1} \sum_{I(x,y) \in p_1} I(x, y) + \frac{1}{N_2} \sum_{I(x,y) \in p_2} I(x, y) \right)$$

while $|\mu - T| > V$ **do** $\mu_{\text{new}} \leftarrow T$

$$p_1(x, y) \leftarrow I(x, y) > \mu_{\text{new}}, p_2(x, y) \leftarrow I(x, y) \leq \mu_{\text{new}}$$

$$T_{\text{new}} \leftarrow \frac{1}{2} \left(\frac{1}{N_1} \sum_{I(x,y) \in p_1} I(x, y) + \frac{1}{N_2} \sum_{I(x,y) \in p_2} I(x, y) \right)$$

end while

$$b(x, y) \leftarrow \begin{cases} 1, & I(x, y) \geq T_{\text{new}} \\ 0, & I(x, y) < T_{\text{new}} \end{cases}$$

Output: $b(x, y)$.

An example of the ROI extraction process applied to both range-time maps and micro-Doppler signatures is shown in Fig. 1.

E. Activity Segmentation

For activity segmentation, the power-weighted centroid of the range-time detections is computed for the each time instant following the ROI extraction process. By calculating the centroid at each time instance, the target trajectory over time is obtained, which is subsequently used to segment the activity phases.

After obtaining the trajectory, the activity phases are segmented using the regression-based model proposed in [10]. The TUG test range-time model is a nonlinear parametric model composed of multiple connected linear segments, with nonlinearity arising at the segment junctions. The model consists of a horizontal segment representing standing up, followed by a positively sloped segment corresponding to forward walking, a horizontal segment for turning, a negatively

sloped segment for the return phase, and a final horizontal segment representing sitting down [10].

The parametric range-time model is defined as follows:

$$f(x) = \begin{cases} y_1, & x \leq b_1, \\ y_1 + m_1(x - b_1), & b_1 < x \leq b_2, \\ y_1 + m_1(b_2 - b_1), & b_2 < x \leq b_3, \\ y_1 + m_1(b_2 - b_1) + m_2(x - b_3), & b_3 < x \leq b_4, \\ y_1 + m_1(b_2 - b_1) + m_2(b_4 - b_3), & x > b_4. \end{cases} \quad (3)$$

The model parameters comprise the connection points (b_1 – b_4), the initial height (y_1), and the slopes of the second and fourth segments (m_1 for the forward phase and m_2 for the return phase), resulting in a total of seven parameters to be estimated for each regression.

The initial values for the regression parameters are determined following the procedure described in [10]. Using these initialisations, the Levenberg–Marquardt algorithm is employed to estimate the activity breakpoints (b_1 – b_4). Based on the resulting time instances, the micro-Doppler signatures are segmented for subsequent spatio-temporal gait analysis.

F. Spatiotemporal Gait Parameter Extraction

The gait envelope is first obtained from the segmented micro-Doppler signatures following ROI extraction, forming the basis for spatio-temporal gait parameter estimation [17]. At each time instant, the maximum and minimum values within the ROI are identified, and the gait envelope is derived from the maximum values.

Two key gait cycle events are identified from the resulting gait envelope for spatio-temporal parameter extraction, namely the toe-off and heel-strike events. The toe off event is defined as the time instant at which the leg related Doppler component departs from near zero velocity, indicating the onset of the swing phase. The heel strike event corresponds to the time instant at which the leg Doppler component returns to near zero velocity, marking the end of the swing phase and foot contact with the ground. An illustration of the gait envelope with the corresponding gait cycle events extracted from the segmented micro-Doppler signature is shown in Fig. 2.

Following the extraction of gait cycle events, a set of spatiotemporal gait parameters, including step time, stride time, cadence, double support time, swing time, and stance

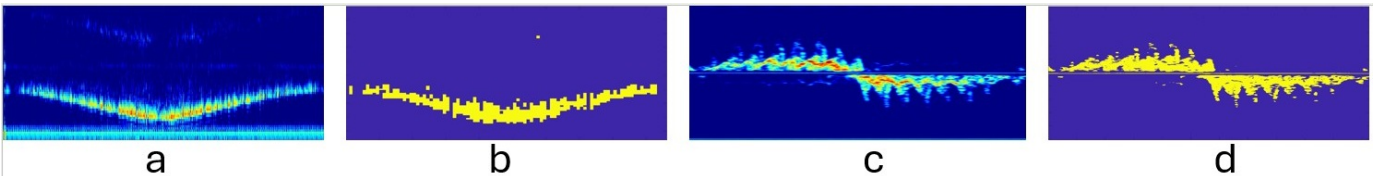


Fig. 1. Region of interest (ROI) extraction: (a) range-time map, (b) ROI-extracted range-time map, (c) micro-Doppler signature, and (d) ROI-extracted micro-Doppler signature.

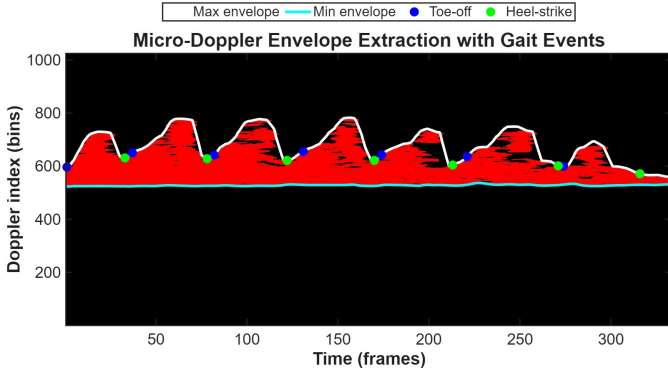


Fig. 2. Micro-Doppler envelope extraction with gait cycle events

time, are derived in accordance with established methods [17], [18].

III. RESULTS AND DISCUSSION

The spatiotemporal gait parameters extracted from the radar measurements were compared against ground truth values obtained from the Vicon MoCap system. The comparison was performed on a per-parameter basis by matching corresponding gait events and temporal segments between the radar and MoCap measurements.

The radar derived spatiotemporal gait parameters were quantitatively evaluated against MoCap measurements using mean absolute error (MAE) and root mean square error (RMSE). Let x_i^r and x_i^m denote the radar estimated and MoCap reference values for the i th gait parameter, respectively, and N be the total number of samples. MAE and RMSE are defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |x_i^r - x_i^m|, \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^r - x_i^m)^2}. \quad (5)$$

In addition, a relative accuracy metric was computed for continuous time and cadence parameters based on the mean absolute percentage error (MAPE), defined as $100 - \text{MAPE}$. This accuracy metric quantifies the relative deviation from the MoCap reference and should not be interpreted as classification accuracy. MAPE based accuracy was not reported for double support due to its percentage based representation. Table I shows that step time and cadence are estimated with high accuracy, whereas stance, swing, and stride times exhibit higher errors, reflecting the inherent challenges of gait event estimation using radar measurements.

The higher errors observed for stance, swing, and stride times are mainly attributed to the indirect estimation of toe-off events from radar micro-Doppler signatures and the accumulation of timing uncertainties over multiple gait cycles. In contrast, step time and cadence rely primarily on heel-strike detection, which can be estimated more robustly from the dominant Doppler components.

Despite these limitations, the achieved accuracy levels indicate that the proposed radar-based approach is suitable for non-contact gait monitoring applications, where relative changes in spatiotemporal parameters are often of greater interest than exact clinical equivalence.

IV. CONCLUSION AND FUTURE DIRECTIONS

This paper investigated the extraction of spatiotemporal gait parameters from radar measurements and their validation against MoCap data. The presented results confirm the feasibility of radar-based gait analysis for non-contact and privacy-preserving monitoring scenarios for early detection of mobility impairments.

Future work will focus on improving the robustness of gait cycle event detection, which is strongly dependent on accurate and noise-free envelope extraction. In this study, an image-based ROI extraction approach was employed; however, this method does not fully preserve micro-Doppler signatures and is not well suited for embedded implementations. To address these limitations, future work will investigate CFAR-based [19] ROI extraction techniques that enable more reliable envelope estimation while remaining computationally efficient for real-time and embedded radar systems.

Another future direction concerns the activity segmentation stage, which currently relies on a regression based model. While effective under controlled conditions, this approach may be less suitable for realistic scenarios where walking durations vary or where individuals exhibit irregular gait patterns, particularly in the presence of early stage mobility impairments. Future work will therefore investigate more adaptive activity segmentation strategies that are robust to variable walking durations and non ideal gait behaviours.

An additional future direction will investigate the use of a super-resolution technique [20] to obtain more detailed micro-Doppler signatures. Enhanced Doppler resolution is expected to improve the representation of limb motion and inter-limb dynamics, thereby enabling more accurate gait cycle event detection and improved spatiotemporal parameter estimation, particularly in challenging or noisy measurement conditions.

Finally, future work will explore the use of multiple receiver antennas, potentially within a MIMO radar configuration, to improve spatial resolution and enable the extraction of richer spatiotemporal gait parameters. Increased spatial diversity is expected to support more reliable target separation and facilitate the analysis of complex gait patterns.

TABLE I
VALIDATION RESULTS OF RADAR DERIVED SPATIOTEMPORAL GAIT PARAMETERS AGAINST MoCAP MEASUREMENTS.

Parameter	Unit	MAE	RMSE	Accuracy (%)
Step time	s	0.0196	0.0317	96.43
Stride time	s	0.1421	0.1726	89.64
Stance time	s	0.0900	0.1135	88.27
Swing time	s	0.0813	0.1086	86.23
Double support	%	9.4965	11.1611	–
Cadence	steps/min	3.7630	6.9958	96.58

V. ACKNOWLEDGEMENT

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REFERENCES

- [1] do Nascimento, Carla Ferreira, et al. "Early identification of older individuals at risk of mobility decline with machine learning." *Archives of Gerontology and Geriatrics* 100 (2022): 104625.
- [2] Podsiadlo, Diane, and Sandra Richardson. "The timed "Up and Go": a test of basic functional mobility for frail elderly persons." *Journal of the American geriatrics Society* 39.2 (1991): 142-148.
- [3] Wang, Fang, et al. "Toward a passive low-cost in-home gait assessment system for older adults." *IEEE journal of biomedical and health informatics* 17.2 (2013): 346-355.
- [4] Kristoffersson, Annica, and Maria Lindén. "A systematic review of wearable sensors for monitoring physical activity." *Sensors* 22.2 (2022): 573.
- [5] Muro-De-La-Herran, Alvaro, Begonya Garcia-Zapirain, and Amaia Mendez-Zorrilla. "Gait analysis methods: An overview of wearable and non-wearable systems, highlighting clinical applications." *Sensors* 14.2 (2014): 3362-3394.
- [6] Le Kernec, Julien, et al. "Radar signal processing for sensing in assisted living: The challenges associated with real-time implementation of emerging algorithms." *IEEE Signal Processing Magazine* 36.4 (2019): 29-41.
- [7] Fioranelli, Francesco, Julien Le Kernec, and Syed Aziz Shah. "Radar for health care: Recognizing human activities and monitoring vital signs." *IEEE Potentials* 38.4 (2019): 16-23.
- [8] Seifert, Ann-Kathrin, Moeness G. Amin, and Abdelhak M. Zoubir. "Toward unobtrusive in-home gait analysis based on radar micro-Doppler signatures." *IEEE Transactions on Biomedical Engineering* 66.9 (2019): 2629-2640.
- [9] NHS England. NHS Long Term Workforce Plan. NHS England, 30 June 2023, www.england.nhs.uk/long-read/nhs-long-term-workforce-plan-2/. Accessed 5 Jan. 2026.
- [10] Bordat, Alexandre, et al. "Towards Automated Fall Risk Assessment: Validation of an FMCW Radar-Based Timed Up and Go Test." *IEEE Transactions on Radar Systems* (2025).
- [11] Li, Xinrong, et al. "Signal processing for TDM MIMO FMCW millimeter-wave radar sensors." *IEEE Access* 9 (2021): 167959-167971.
- [12] Cohen, Leon. "Time-frequency analysis." Englewood Cliffs (1995).
- [13] Ullmann, Ingrid, et al. "A survey on radar-based continuous human activity recognition." *IEEE Journal of Microwaves* 3.3 (2023): 938-950.
- [14] Shrestha, Aman, et al. "Continuous human activity classification from FMCW radar with Bi-LSTM networks." *IEEE Sensors Journal* 20.22 (2020): 13607-13619.
- [15] Li, Zhenghui, et al. "Radar-based human activity recognition with adaptive thresholding towards resource constrained platforms." *Scientific Reports* 13.1 (2023): 3473.
- [16] Akaydin, Abdullah, et al. "Enhancing human activity recognition with iterative adaptive approach in assisted living." *IEEE Sensors Journal* (2025).
- [17] Seifert, Ann-Kathrin. Signal processing for radar-based gait analysis. Diss. 2020.
- [18] Hadjipanayi, Charalambos, et al. "Remote gait analysis using ultra-wideband radar technology based on joint range-doppler-time representation." *IEEE Transactions on Biomedical Engineering* 71.10 (2024): 2854-2865.
- [19] Akaydin, Abdullah, and Mucahit Kani Üner. "Comparative Performance Analysis of Constant False Alarm Rate Processors Based on Order Statistics." 2019 11th International Conference on Electrical and Electronics Engineering (ELECO). IEEE, 2019.
- [20] Akaydin, Abdullah, and Julien Le Kernec. "Micro-Doppler Super-Resolution Using Iterative Adaptive Approach." 2024 International Radar Conference (RADAR). IEEE, 2024.